

Aufgabenstellung Projektarbeit

Hochschule Karlsruhe
University of
Applied Sciences
Fakultät für
Maschinenbau und
Mechatronik



Projekttitel: Simulation-Based Reinforcement Learning for Autonomous RC Car driving

Background: This student project develops a Reinforcement Learning agent that autonomously drives a Carrera Hybrid miniature racing car around a track, optimizing lap time while keeping the vehicle within the track boundaries. Since training on a physical track is impractical, the agent is developed and trained entirely in simulation. To reduce iteration time on fundamental design decisions, such as which observations the agent receives (single image, image sequence, image plus previous actions), the core architecture is first prototyped in a simplified 2D Python environment before being transferred to the more realistic NVIDIA Isaac Sim / Isaac Lab setup.

The project will include the following goals or outcomes:

- Develop a lightweight 2D Python environment to prototype and evaluate fundamental RL design choices, including observation type, history length, and reward signal structure.
- Build a physics-based simulation environment in Isaac Sim / Isaac Lab, including a simplified car articulation and an oval track replicating the dimensions of a real Carrera Hybrid circuit.
- Transfer the validated RL architecture into an Isaac Lab `DirectREnv` with continuous action space (steering and throttle) and a reward function encouraging fast and stable driving along the track centerline.
- Train a Proximal Policy Optimization (PPO) agent capable of completing the track autonomously without leaving its boundaries.
- Evaluate the trained policy using quantitative metrics such as lap time, off-track frequency, and average distance from the centerline.

Dokumentation

Bericht, Präsentation, Poster

Bearbeitung im: SS 2026	Zielgruppe: Robotik und KI i.d. Prod. Master	Projekt-Code: 26ss_BH_CarreraRC_RL
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